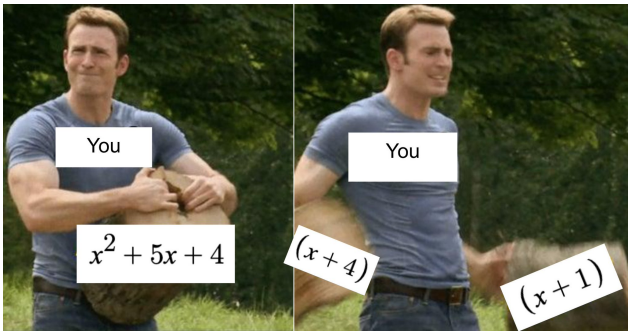

Grade 12 Advanced Functions
Factor Theorem Key

First Name: _____**KEY**_____

Last Name: _____

Directions:

- Please answer to 2 decimal points
- The test is designed to be completed in 75 minutes

For grading use only

Page:	2	3	4	5	Total
Points:	10	8	8	4	30
Score:					

Multiple Choice (6 marks)

1. (1 point) Which value of a makes $(x - a)$ a factor of $f(x) = x^3 + 2x^2 - x - 2$?
   A. -2
   B. -1
   C. 0
   D. 1
2. (1 point) If $f(x) = x^3 - 7x + 6$, which of the following is NOT a factor?
   A. $(x - 1)$
   B. $(x - 2)$
   C. $(x + 3)$
   D. $(x + 2)$
3. (1 point) What is the remainder when $f(x) = 2x^3 - 5x^2 + 7x - 1$ is divided by $(x - 2)$?
   A. -1
   B. 0
   C. 7
   D. 9
4. (1 point) Which binomial is a factor of $f(x) = x^3 + 3x^2 + 2x$?
   A. $(x - 2)$
   B. $(x - 1)$
   C. $(x + 2)$
   D. $(x + 3)$
5. (1 point) If $f(x) = x^3 - x^2 - 4x + 4$, which of the following is a root?
   A. -2
   B. 0
   C. 2
   D. 1
6. (1 point) Which value of k makes $(x + 3)$ a factor of $f(x) = 2x^3 + kx^2 + 5x - 3$?
   A. -3
   B. 0
   C. 6
   D. 8

Long Answer (24 marks)

7. **Factor Theorem Application**
   (a) (4 points) Is $(x-2)$ a factor of $f(x) = x^3 - 3x^2 + 4$? Justify your answer using the Factor Theorem.

Solution: By the Factor Theorem, $(x-2)$ is a factor of $f(x)$ if $f(2)=0$.

Calculate $f(2)$:

$$f(2) = (2)^3 - 3(2)^2 + 4 = 8 - 12 + 4 = 0$$

Since $f(2)=0$, $(x-2)$ is a factor of $f(x)$.

To confirm, perform synthetic division with root 2:

Coefficients: $1 \quad -3 \quad 0 \quad 4$

Carry down 1: 1 Multiply by 2, add to -3: -1 Multiply by 2, add to 0: -2 Multiply by 2, add to 4: 0

So, the quotient is $x^2 - x - 2$, and the remainder is 0.

Therefore,

$$f(x) = (x-2)(x^2 - x - 2)$$

Further factor $x^2 - x - 2 = (x-2)(x+1)$.

Final factorization:

$$f(x) = (x-2)^2(x+1)$$

8. **Finding a Coefficient for a Factor**

   (a) (4 points) Find a value of k such that $(x+3)$ is a factor of $f(x) = 2x^3 + kx^2 + 5x - 3$.

Solution: By the Factor Theorem, $(x+3)$ is a factor if $f(-3)=0$.

Calculate:

$$f(-3) = 2(-3)^3 + k(-3)^2 + 5(-3) - 3 = 2(-27) + k(9) - 15 - 3 = -54 + 9k - 18$$

Combine constants:

$$-54 - 15 - 3 = -72$$

So,

$$f(-3) = 9k - 72$$

Set $f(-3)=0$:

$$9k - 72 = 0 \implies 9k = 72 \implies k = 8$$

Therefore, $k=8$ makes $(x+3)$ a factor of $f(x)$.

9. **Factoring a Cubic Polynomial**

   (a) (4 points) Factor $f(x) = x^3 - 7x + 6$ completely.

Solution: Try integer roots from -3 to 3:

$$f(1) = 1 - 7 + 6 = 0 \implies (x-1) \text{ is a factor}$$

Use synthetic division to divide by $(x-1)$:

Coefficients: $1 \quad 0 \quad -7 \quad 6$

Carry down 1: 1 Multiply by 1, add to 0: 1 Multiply by 1, add to −7: −6 Multiply by 1, add to 6: 0

So, $f(x) = (x-1)(x^2+x-6)$

Now factor x^2+x-6**: Find two numbers that multiply to −6 and add to 1: 3 and −2.**

So,

$$x^2+x-6 = (x+3)(x-2)$$

Final factorization:

$$f(x) = (x-1)(x+3)(x-2)$$

Check all roots:

$$f(-3) = (-3)^3 - 7(-3) + 6 = -27 + 21 + 6 = 0$$

$$f(2) = 8 - 14 + 6 = 0$$

All roots are valid.

10. **Determining a Root**

(a) (4 points) Determine if $x = -2$ is a root of $f(x) = x^3 + 3x^2 + 2x$.

Solution: Calculate $f(-2)$**:**

$$f(-2) = (-2)^3 + 3(-2)^2 + 2(-2) = -8 + 12 - 4 = 0$$

Since $f(-2) = 0$**,** $x = -2$ **is a root and** $(x+2)$ **is a factor.**

To factor further, perform synthetic division with root −2:

Coefficients: 1 3 2 0

Carry down 1: 1 Multiply by −2, add to 3: 1 Multiply by −2, add to 2: 0 Multiply by −2, add to 0: 0

So, $f(x) = (x+2)(x^2+x)$

Factor $x^2 + x = x(x+1)$

Final factorization:

$$f(x) = (x+2)x(x+1)$$

Roots: $x = -2, 0, -1$

11. **Remainder Theorem**

(a) (4 points) What is the remainder when $f(x) = 2x^3 - 5x^2 + 7x - 1$ is divided by $(x-2)$?

Solution: By the Remainder Theorem, the remainder is $f(2)$**:**

$$f(2) = 2(8) - 5(4) + 7(2) - 1 = 16 - 20 + 14 - 1 = 9$$

So, the remainder is 9.

To confirm, perform synthetic division with root 2:

Coefficients: $2 \quad -5 \quad 7 \quad -1$

Carry down 2**:** 2 **Multiply by** 2**, add to** -5**:** -1 **Multiply by** 2**, add to** 7**:** 5 **Multiply by** 2**, add to** -1**:** 9

So, the remainder is 9**.**

12. **Factoring with Integer Roots**

(a) (4 points) Factor $f(x) = x^3 + x^2 - 4x - 4$ completely.

Solution: Try integer roots from -3 **to** 3**:**

$$f(-1) = (-1)^3 + (-1)^2 - 4(-1) - 4 = -1 + 1 + 4 - 4 = 0$$

So $(x+1)$ **is a factor.**

Use synthetic division to divide by $(x+1)$**:**

Coefficients: $1 \quad 1 \quad -4 \quad -4$

Carry down 1**:** 1 **Multiply by** -1**, add to** 1**:** 0 **Multiply by** -1**, add to** -4**:** -4 **Multiply by** -1**, add to** -4**:** 0

So, $f(x) = (x+1)(x^2-4)$

Factor $x^2 - 4 = (x-2)(x+2)$

Final factorization:

$$f(x) = (x+1)(x-2)(x+2)$$

Check all roots:

$$f(2) = 8 + 4 - 8 - 4 = 0$$

$$f(-2) = -8 + 4 + 8 - 4 = 0$$

All roots are valid.